



Exploratory Data Analysis

Lab Activity 1: Dataset Collection and Exploratory Data Analysis (EDA)

Instruction

What are the first steps to take when you start a project or get ahold of a new data set?

In a real data science project, you will typically have a question or idea brought to you: How are people interacting with the new version of my software? How is weather in Echague changing in the last decade? What type of people are most likely to enroll in Data Science Analytics? You will sometimes be given a dataset when asked one of these questions, or more often, a general description of the data you *could* get if you talked to the right people or interacted with the right software systems.

Choose one of the following two options:

1. A data set from the awesome public datasets() page
2. your own data

Complete the following tasks:

1. Identify team members (2 members).
2. Formulate a broad research question.
3. Perform some exploratory data analysis. Describe the steps that you have taken to arrive at your conclusions.
4. Develop at least one visualization (**you can have more**) that tells a story about your more specific research question/hypothesis. Note: the story may very well be something along the lines of “we thought variable X would affect measure Y in way Z, but the evidence does not support that.”
5. Particularly important questions about how it was collected include whether it is a sample of a larger data set, and, if so, how the sampling was done? Randomly? All cases during a specific time frame? All data for a selected set of users? Answers to such questions strongly impact the conclusions you will be able to draw from the data.
6. The graphics in this one are a bit more developed than you would really see in an exploratory analysis, but I think the progression of visualizations is interesting and follows an exploratory story.

Deliverables:

1. **Documentation of the steps and results achieved**
2. **Python/R code-base showing process of exploration**

Grading Rubric (100 points):

- Dataset Collection & Description – 15 pts
- Pre-processing & Cleaning – 25 pts
- Descriptive Statistics & Visualization – 25 pts
- Outlier & Missing Data Analysis – 15 pts
- Insights & Hypotheses – 20 pts



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Example:

Here are data about flight delays from Kaggle(<https://www.kaggle.com/usdot/flight-delays>). There are three tables of data:

```
airlines <- read.csv("https://www.macalester.edu/~dshuman1/data/112/flights/airlines.csv")
airports <- read.csv("https://www.macalester.edu/~dshuman1/data/112/flights/airports.csv")
flights <- read.csv("https://www.macalester.edu/~dshuman1/data/112/flights/flights_jan_jul_sample2.csv")
```

Note that the full set of flight data has more than 5.8 million flights. To start, I have given you a subset that includes all flights in the first 15 days of January 2015 and in the first 15 days of July 2015. If at the end you wish to try your analysis on the whole data set, you can download the original .csv file from the Kaggle page and substitute a link to it in the code chunk above.

There is a ton of data here, and it can be easy to be overwhelmed. We are going to focus our exploration a bit by considering the following broad research question: **which flights are most likely to be delayed?**

Exercise 15.1 (Data source) Where does this data come from? Who collected it?

Exercise 15.2 (Explore codebook) This data set comes with a codebook on the Kaggle site. Have a look at the codebook to understand which variables are contained in each of the three tables. For example, what are the levels of CANCELLATION_REASON and what do they mean? What is the unit of observation for each table?

Exercise 15.3 (Possible joins) What variables link the three tables? How could you join data from one table to another?

Exercise 15.4 (Visualize and describe the data) Use some univariate and bivariate visualizations to start to explore the questions mentioned above:

- What do you see that is interesting?
- Which values are most common or unusual (outliers)?
- Is there a lot of missing data?
- What type of variation occurs within the individual variables?
- What might be causing the interesting findings?
- How could you figure out whether your ideas are correct?

Exercise 15.5 (Formulate a specific research question) Based on your preliminary visualizations and exploration of the data, formulate a more specific research question/hypothesis/conclusion within this broad area of understanding the causes of flight delays.